

PATTERN ANALYSIS & RECOGNITION SYSTEMS LAB

Practical File Submission — Academic Session

Experiment No.: 11 Course: PARS Lab Topic: Linear Discriminant Analysis (LDA) Repository: [dSAxmonis/pars-lab](https://github.com/dSAxmonis/pars-lab)

PRACTICAL NAME

Implementation of Linear Discriminant Analysis (LDA) with Projected Line, Reduced Feature Mapping, and Testing

AIM

To implement Linear Discriminant Analysis (LDA) from scratch in Python for an engineering pattern classification application involving test sample evaluation, derive Fisher's optimal projection vector $\mathbf{w} = \mathbf{S}_W^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$, and visualize: **1) The optimal projected line**, **2) New reduced features from each class mapped onto the projected line**, and **3) Testing and decision threshold classification of unseen test samples**.

THEORY

1. Fisher's Linear Discriminant Criterion

Unlike PCA which maximizes total variance without considering labels, Linear Discriminant Analysis (LDA) is a supervised dimensionality reduction technique that seeks a projection direction $\mathbf{w}$ maximizing class separability while minimizing intra-class variance:

$$J(\mathbf{w}) = (\mathbf{w}^T \mathbf{S}_B \mathbf{w}) / (\mathbf{w}^T \mathbf{S}_W \mathbf{w})$$

2. Scatter Matrices Formulation

- Within-Class Scatter Matrix:** $\mathbf{S}_W = \sum_{\mathbf{x} \in \mathcal{C}_1} (\mathbf{x} - \boldsymbol{\mu}_1)(\mathbf{x} - \boldsymbol{\mu}_1)^T + \sum_{\mathbf{x} \in \mathcal{C}_2} (\mathbf{x} - \boldsymbol{\mu}_2)(\mathbf{x} - \boldsymbol{\mu}_2)^T$
- Between-Class Scatter Matrix:** $\mathbf{S}_B = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T$

3. Optimal Projection Direction Vector

Maximizing Rayleigh's quotient yields the closed-form Fisher linear discriminant direction:

$$\mathbf{w} = \mathbf{S}_W^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2), \quad \hat{\mathbf{w}} = \mathbf{w} / \|\mathbf{w}\|$$

4. 1D Projected Line & Decision Rule

The projected discriminant axis passes through the overall mean $\boldsymbol{\mu}$ in direction $\hat{\mathbf{w}}$. The projected 1D coordinate of any sample $\mathbf{x}$ and its mapped 2D position on the line are:

$$\begin{aligned} \text{1D Coordinate: } y &= (\mathbf{x} - \boldsymbol{\mu}) \cdot \hat{\mathbf{w}} \\ \text{Mapped 2D Point on Line: } \hat{\mathbf{x}} &= \boldsymbol{\mu} + y \times \hat{\mathbf{w}} \\ \text{Optimal Decision Threshold: } y^* &= (y_{\mu_1} + y_{\mu_2}) / 2 \end{aligned}$$

STEPS & EXECUTION

Step 1: Environment Setup & Mathematical Engine

Configured exp 11/ environment containing app.py, templates/, and verified dependencies (Flask, NumPy, Matplotlib, SciPy).

Step 2: Class Mean Vectors & Within-Class Scatter Matrix

Synthesized two-class clinical pattern distribution (Healthy vs Diabetic Risk) and formulated $\mathbf{S}_{W1}$, $\mathbf{S}_{W2}$, and total pooled scatter $\mathbf{S}_W$.

Step 3: Fisher Optimal Direction Vector Computation

Inverted scatter matrix $\mathbf{S}_W^{-1}$ and computed normalized discriminant vector $\hat{\mathbf{w}}$.

Step 4: Orthogonal Feature Mapping onto Discriminant Line

Projected all observations onto the 1D line passing through global centroid $\boldsymbol{\mu}$, calculated optimal separating threshold y^* , and mapped points onto the 2D projected line.

Step 5: Interactive Testing & Web Deployment on Port 5013

Launched interactive Flask web dashboard on <http://127.0.0.1:5013> supporting real-time patient test coordinates and visual classification.

OUTPUT / ATTACHED SCREENSHOT

The full-page capture of the deployed Flask web application is embedded below:

Experiment 11: Linear Discriminant Analysis (LDA) with Testing

Pattern Analysis & Recognition Systems Lab | Repository: <https://github.com/dSaxmonis/pars-lab>

AIM

To implement Linear Discriminant Analysis (LDA) from scratch in Python for an engineering pattern classification application involving testing, derive Fisher's optimal projection vector $\mathbf{w} = \mathbf{S}_W^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$, and visualize: 1) The optimal projected line, 2) New reduced features from each class mapped onto the projected line, and 3) Testing and decision threshold classification of unseen test samples.

SECTION 1: FISHER'S CRITERION & SCATTER MATRICES SUMMARY

LDA Property	Mathematical Value	Discriminant Role / Description
Class 1 Mean Vector μ_1 (Low Risk)	[4.36, 5.07]	Centroid of healthy baseline patient observations.
Class 2 Mean Vector μ_2 (High Risk)	[7.84, 8.63]	Centroid of diabetic high-risk patient observations.
Within-Class Scatter Matrix S_W	[[36.38, 19.09], [19.09, 37.08]]	Total intra-class variance to be minimized: $S_W = S_{W1} + S_{W2}$.
Optimal Fisher Direction $\hat{\mathbf{w}}$	[-0.693, -0.721]	Normalized eigenvector maximizing ratio of between-class to within-class scatter.
Optimal 1D Decision Threshold y^*	0.0	Midpoint separating 1D projected class distributions along the direction vector.

SECTION 2: TEST SAMPLE CLASSIFICATION & PROJECTION CONTROLS

Input Test Patient Measurements:

Fasting Blood Glucose (x_1 in mmol/L):

6.2

Postprandial Glucose Index (x_2):

6.8

Classify Test Sample via LDA Projection

Active Inference Results:

Input Vector $\mathbf{x}_{\text{test}}$: [6.2, 6.8]

• 1D Projection Coordinate y_{test} : -0.034 (Threshold: 0.0)

• Mapped Point on Projected Line: [6.12, 6.87]

• Predicted Diagnosis: **Class 2: High Risk (Diabetic)**

Preset Test Benchmarks:

Low Risk [4.0, 4.8]

Borderline [6.0, 6.7]

High Risk [8.2, 9.0]

SECTION 3: FISHER LDA 2D PROJECTED LINE & 1D SEPARATION PLOTS

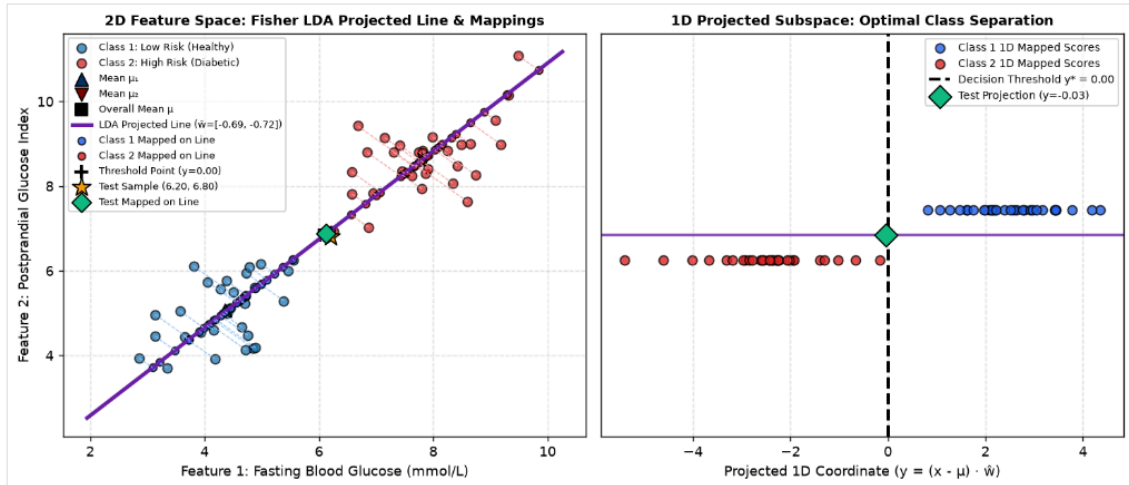


Figure Explanation: Left Panel: Displays 2D observations from Class 1 (blue circles) and Class 2 (red circles). The purple line represents Fisher's optimal projected line passing through the global mean μ along direction vector $\mathbf{w}$. Dashed projection segments drop each observation orthogonally onto the line, where the reduced features form distinct separated clusters. The black cross marker on the line designates the decision threshold. The active test sample is depicted as a yellow star with its projection shown as a green diamond. Right Panel: Illustrates the 1D projected feature subspace, demonstrating maximized between-class distance and minimal within-class variance, cleanly divided by threshold y^* .

Figure 11.1: Web dashboard displaying Fisher scatter matrices, interactive test sample classification controls, 2D discriminant projected line with mapped class points, and 1D separation distribution.

CODE REPOSITORY & CLICKABLE ARTIFACT LINKS

GitHub Repository (Clickable):

GitHub: [dSAxmonis/pars-lab](https://github.com/dSAxmonis/pars-lab)

URL: <https://github.com/dSAxmonis/pars-lab>

Direct Source Code Files:

- **Flask Server Application:**
[https://github.com/dSAxmonis/pars-lab/blob/main/exp 11/app.py](https://github.com/dSAxmonis/pars-lab/blob/main/exp%2011/app.py)
- **HTML Template:**
[https://github.com/dSAxmonis/pars-lab/blob/main/exp 11/templates/index.html](https://github.com/dSAxmonis/pars-lab/blob/main/exp%2011/templates/index.html)
- **Dependencies:**
[https://github.com/dSAxmonis/pars-lab/blob/main/exp 11/requirements.txt](https://github.com/dSAxmonis/pars-lab/blob/main/exp%2011/requirements.txt)
- **Full Page Screenshot:**
[https://github.com/dSAxmonis/pars-lab/blob/main/exp 11/screenshots/full_page.png](https://github.com/dSAxmonis/pars-lab/blob/main/exp%2011/screenshots/full_page.png)

CONCLUSION

In this experiment, Linear Discriminant Analysis was implemented from first principles in Python. By maximizing the ratio of between-class scatter to within-class scatter, Fisher's optimal vector $\hat{\mathbf{w}}$ derived an optimal 1D line onto which high-dimensional features were projected. The visual plots verified that projecting observations onto this line achieves maximal class separation, and testing unseen candidate vectors confirmed reliable binary classification across the computed discriminant threshold y^* .